Printact.co

Academy of Advanced Skills



Dr. Data Ng

- Physicist, Biohacker, Educator & Maker in Asia.
- Winner / Participant / Organizer of 30+ hackathons and 40+ startup weekends in HK, SG, CN and TW
- PhD in Physics and BSc (Majors: Physics, Finance; Minor: Astronomy) at HKU
- MInstP - Member of Institute of Physics, UK
- First Qiskit Advocate appointed by IBM
- Vice-president of Cyberport Startup Alumni Association (CSAA)

Biotech

Engineering

Computation

0x2A Science

Maker Academy

DO SOMETHING THAT MATTER

DeepTech School

http://0x2A.science

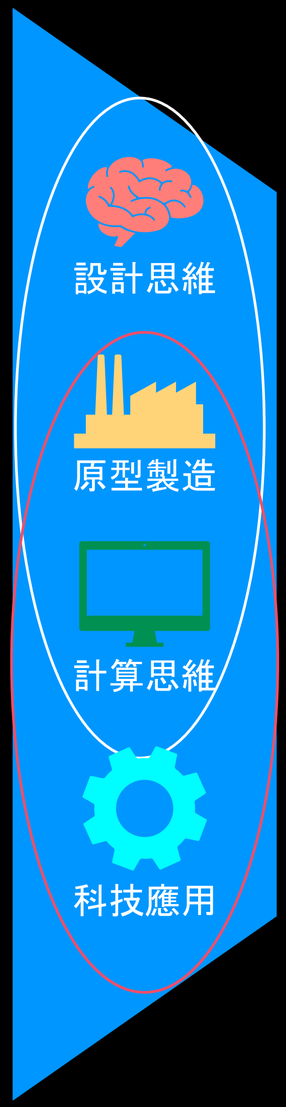
http://makertech.academy

http://Deeptech.school

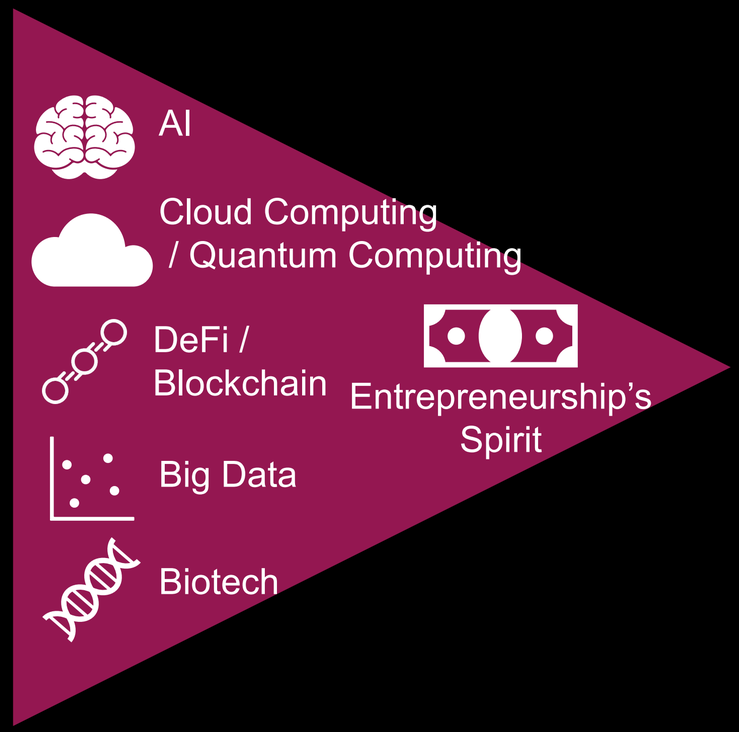
STEM Teaching Approach from University Research



UN SDGs



Basic Skills



Problem Based Learning
(PBL)

Problem-based Learning:
4 pillars of knowledge fields + 5 fields



Basic Skills

Essential skills of Innovation & STEM



Design Thinking



Prototyping



Computational thinking



Applied Tech.



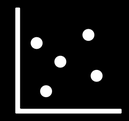
AI



DeFi / Blockchain



Cloud Computing / Quantum Computing



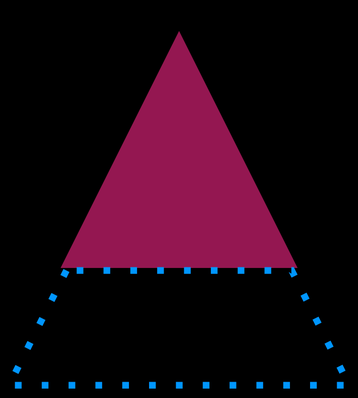
Big Data



Biotech



Entrepreneurship Spirit



Project/ Problem-based Learning (PBL)

Application & Problem Driven

STEM 學科目標:
包含各分科課程的特質和重要元素

STEM+:
融入STEM 以外的科目

設計思維	認知範疇* (認知過程)		STEM 學科範疇目標				其它學科範疇目標
			S 科學 (相關學科:)	T 科技 (相關學科:)	E 工程 (相關學科:)	M 數學 (相關學科:)	其他科目
同理心/需求定義	記憶	事實					
	理解	概念					
		過程					
創意動腦	應用	概念					
		過程					
測試	分析	概念 / 過程					
	評鑑	概念 / 過程					
原型/落實	創造	概念 / 過程					
	後設認知範疇**		指個人對一般學習、思考和解難策略，以及對進行不同任務所須運用的不同策略的了解。				
	廿一世紀技能範疇		指6Cs - 創造力與想像力、批判思維與解難能力、溝通、合作、品格教育與世界公民意識。				
	情意範疇						
	與學科相關的態度		對學習STEM的興趣和自信心，是否有興趣修讀與STEM相關的課程或於將來從事STEM相關職業等。				
	對 STEM 的態度		各STEM 學科本身所蘊含的精神和價值，如科學精神、尊重證據、持續優化、迎難而上、力求精準和精益求精等態度。				

縱向聯繫:
強調不同認知目標層次和知識類型之間的縱向聯繫

橫向聯繫:
突顯不同學科範疇目標之間的橫向聯繫

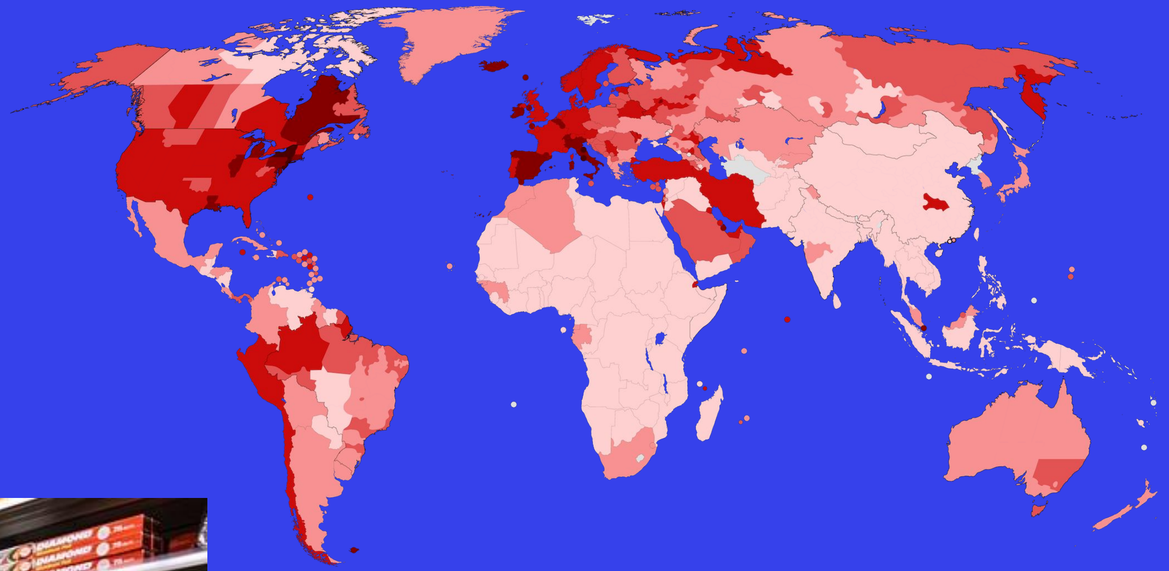
*認知教學目標分類層階 (Bloom's taxonomy) :六個認知目標層次，分別為知識、理解、應用、分析、綜合和評鑑

Biotech

2A Science

http://0x2A.science

COVID-19







Lack of Biotech Knowledge

>60%

biology students have no experience in hands-on experiments



Problem:



Can’t afford expensive bulky biotech equipment

USD$50k for biotech equipment in a science lab



No laboratory in schools

<20% of USA schools own a proper science laboratory



Lack of relevant science background teachers

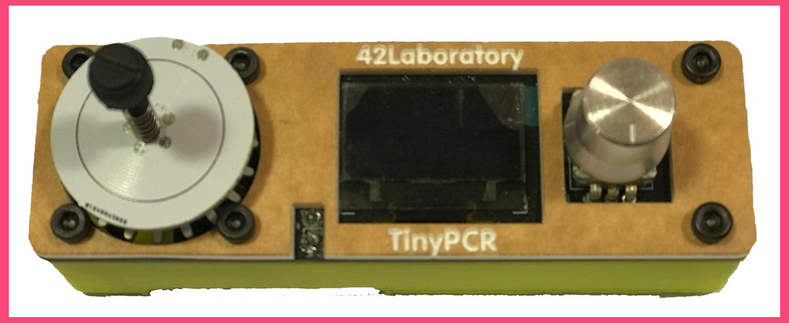
<50% school science teachers with science-related degree



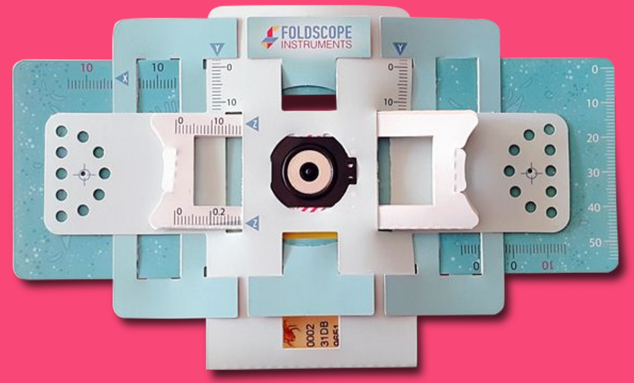
Large student-to-teacher ratio

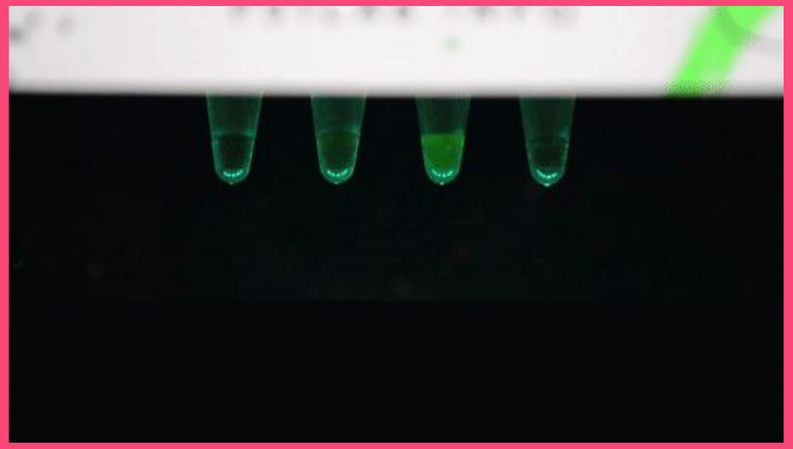
~ 31 Students per class

Advanced Learners













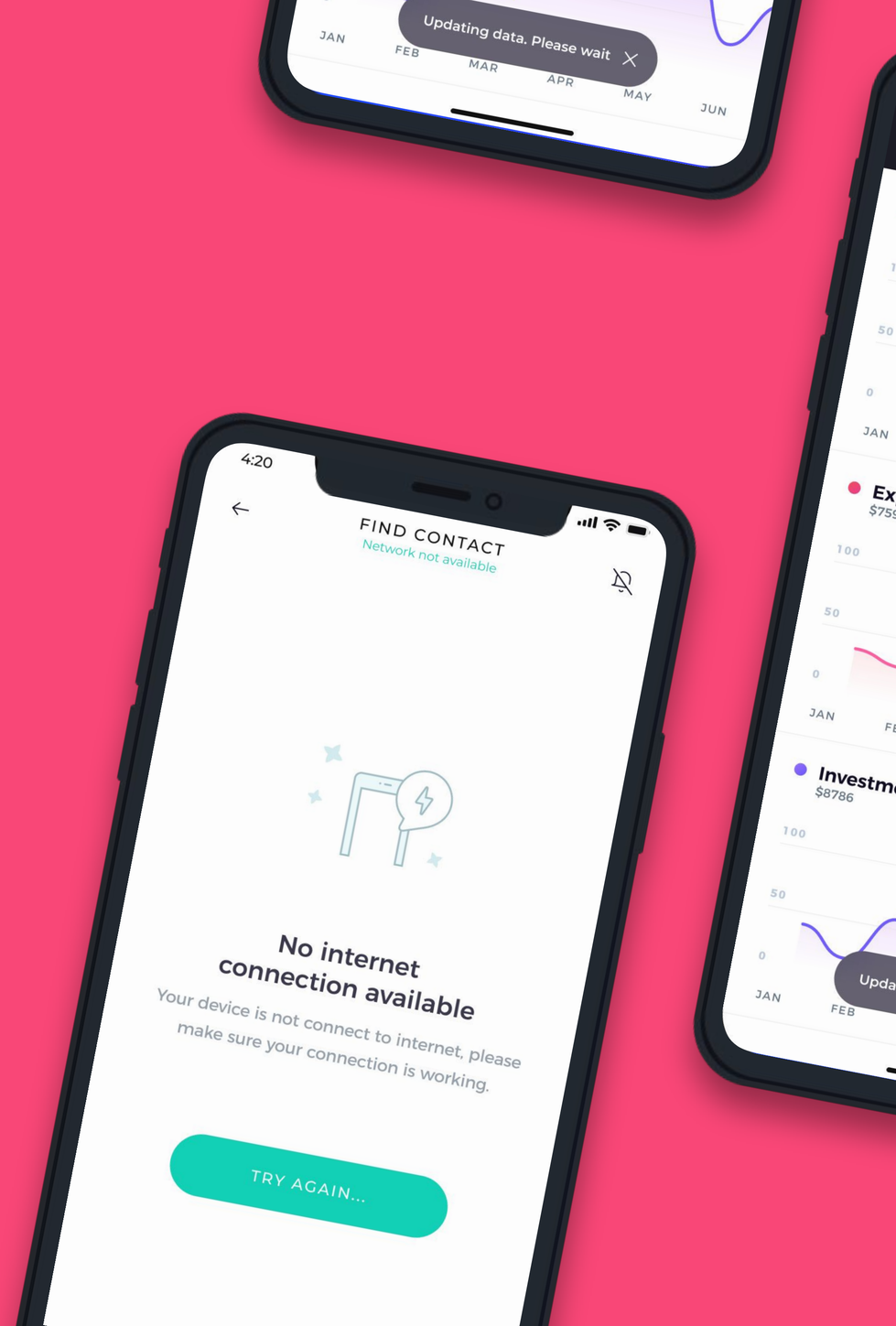
For Beginners

- Booklets for Schools (B2B)
- Teaching Materials
- Learning Management System (B2B & B2C)

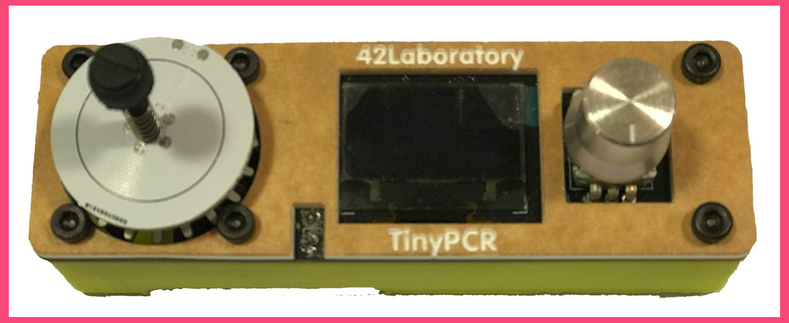


Advanced Learners

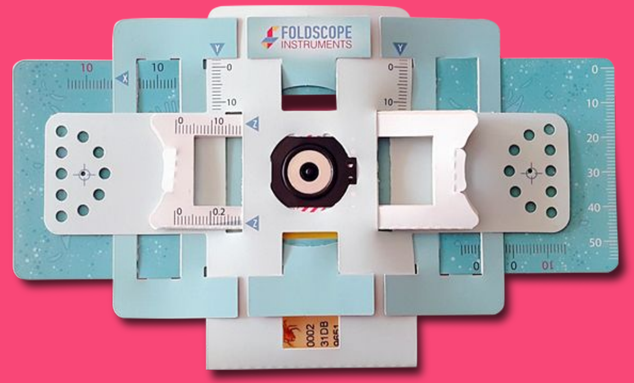
- 6 - infinity y.o.
- Teaching Assistant App
- Freemium
- Monthly Subscription for premium content + support

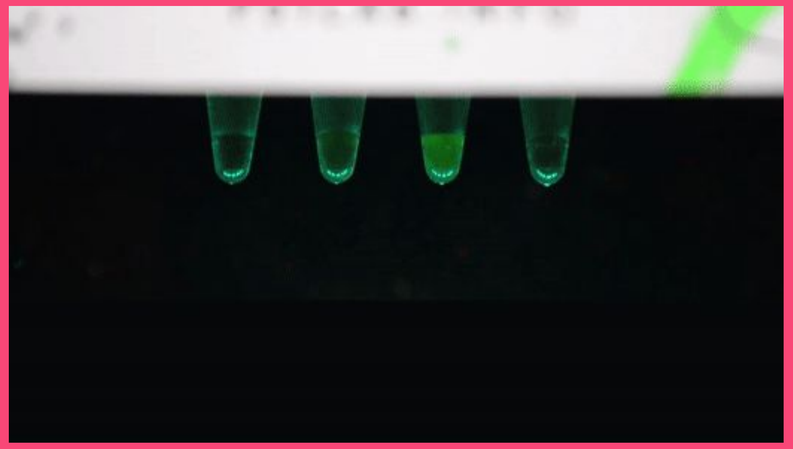


Advanced Learners













Traditional Biotech machine
傳統生物科技儀器

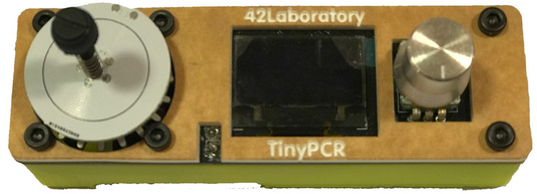
Beyond Dimension

- 13+ years old 歲以上
- 1% traditional equip.
  傳統生物儀器的1%價格
- 1st Single PCB Heating Element
  首創單一發熱原件PCB
- Palm-size: 60g, 10cm 手掌大小
- USB-C Powered 移動電池驅動 PCR

1st Generation
第一代
70cm, 700g

2nd Generation
第二代
10cm

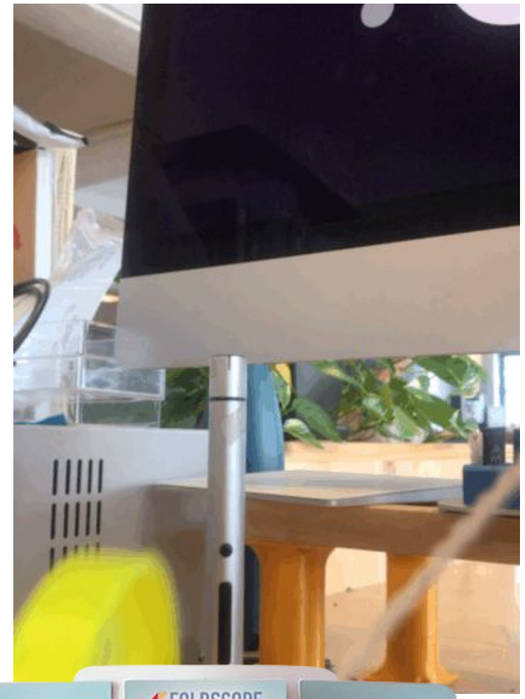
Tiny PCR



Beyond Power

- Hand-powered Centrifuge
  手動離心機
  (6k rpm)
- Paper Microscope
  紙顯微鏡
  (140X, 2 micron)









Beyond Power

Comparing w/. traditional biotech machine:

對比傳統生物科技儀器,

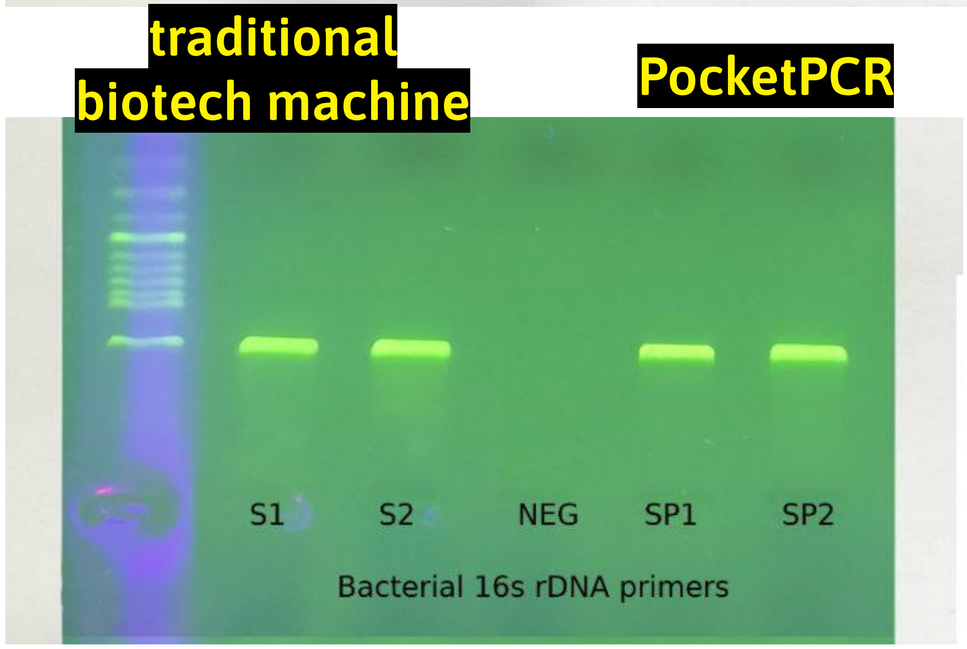
- Same accuracy

  同樣精度

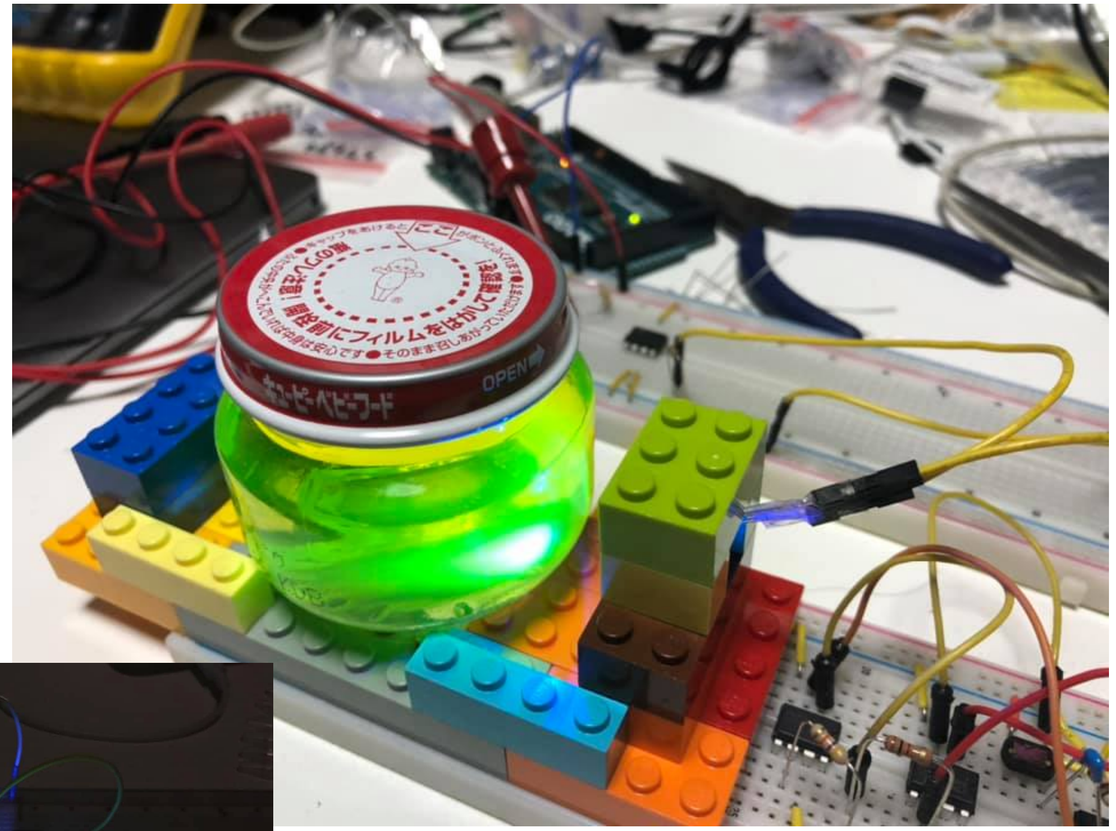
- 10% size & weight

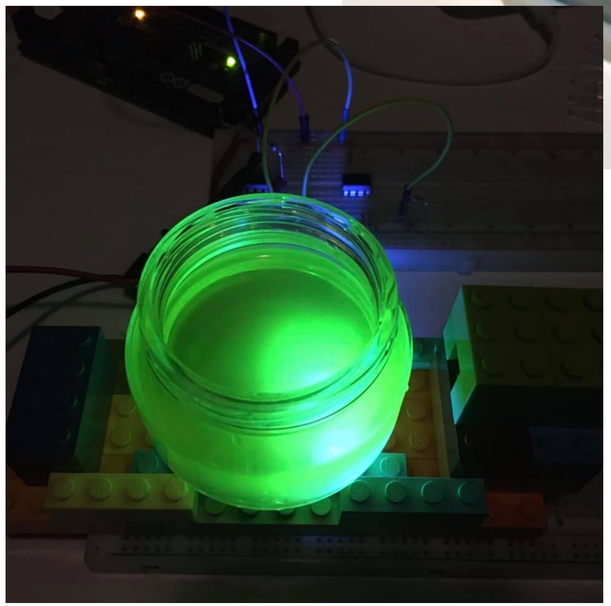
  十分之一大小及重量

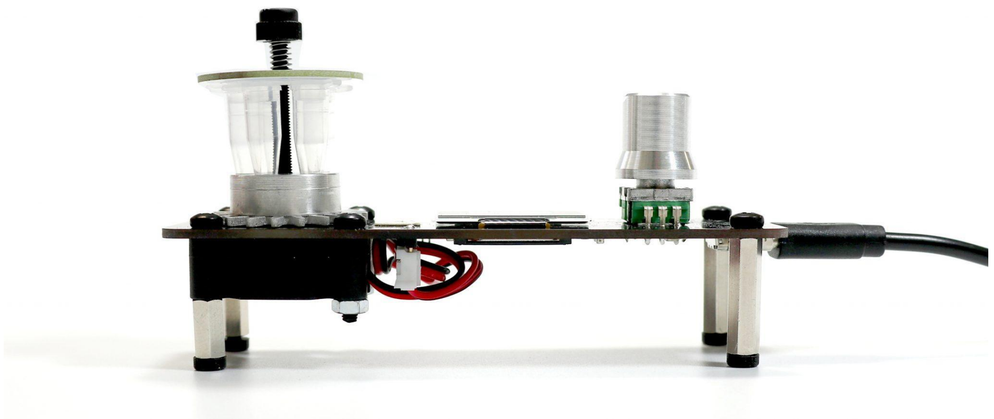




COVID-19 Detection (LAMP)







- Using LAMP: 30 - 45 mins obtained the result

- Using traditional PCR: 2 - 3 hours

PCR多聚酶鏈式反應

1. Extraction
DNA from organism
(e.g. Banana)

由生物提取基因
（如香蕉）



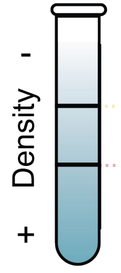




2. Purification
by Centrifuge
to separate DNA
from the mixture

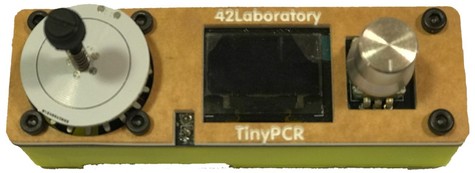
由離心機提純基因

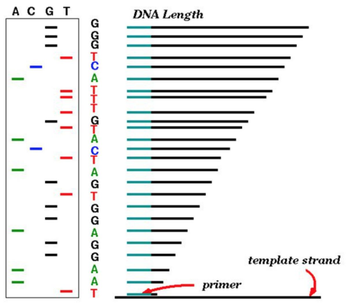




3. PCR
(Polymerase
chain reaction)
to increase the copy
of DNA

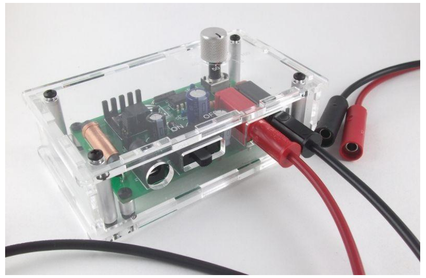
由PCR 增加基因數目





4. Sequencing
DNA fragment by
weight in Gel Tank

凝膠電泳中排序基因



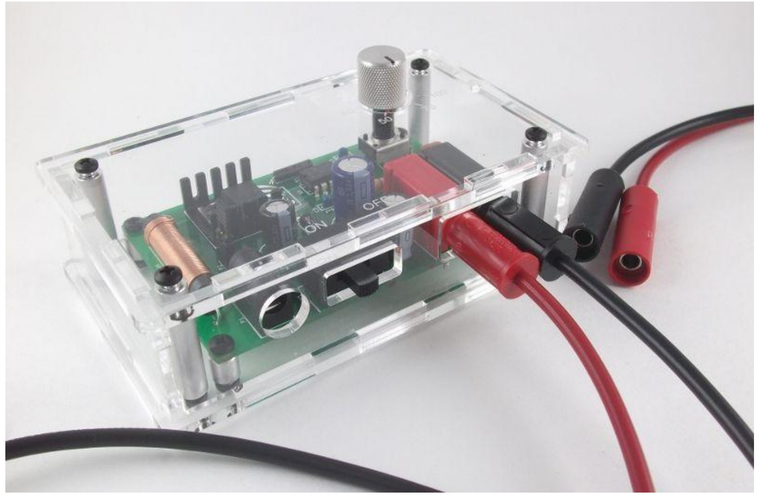


5. Illustration
DNA sequence by
Transilluminator

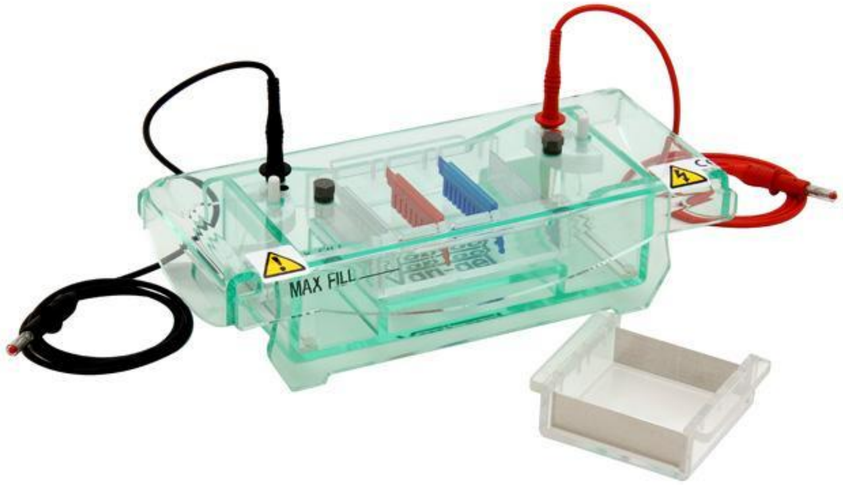
用照膠儀顯影基因排序





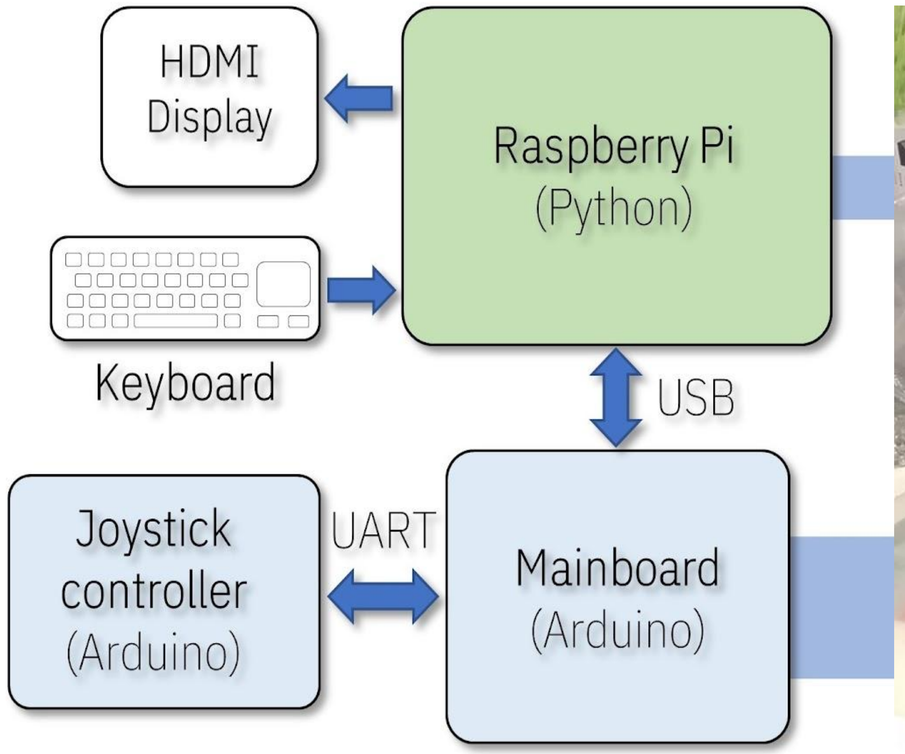

ON
OFF


IO Rodeo
UV-transilluminator


MAX FILL


IO Rodeo
UV-transilluminator

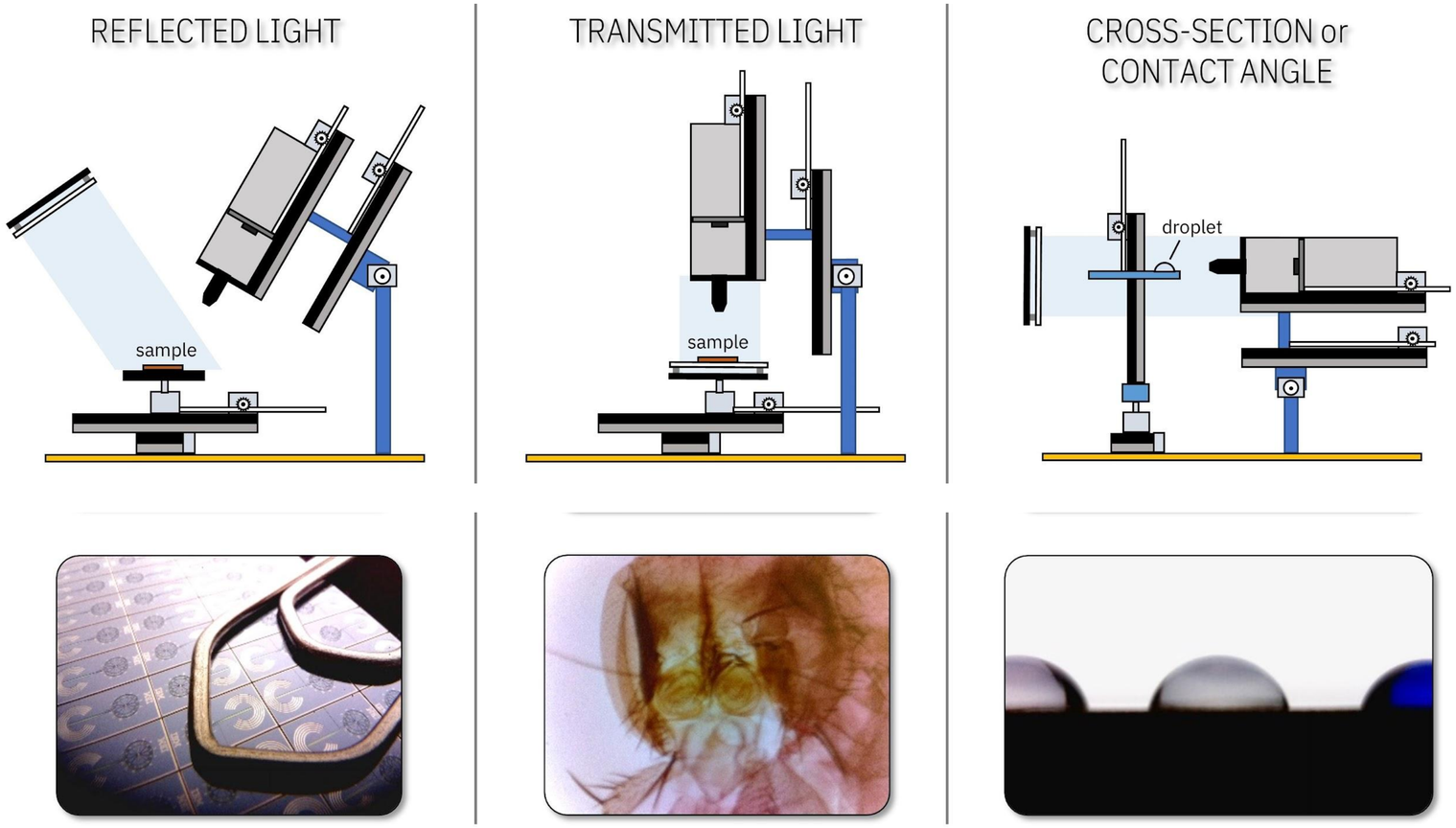
AIoT Microscope





AI(Computer vision) + IoT Connected Microscope

AIoT Microscope



AI(Computer vision) + IoT Connected Microscope

AIoT Microscope



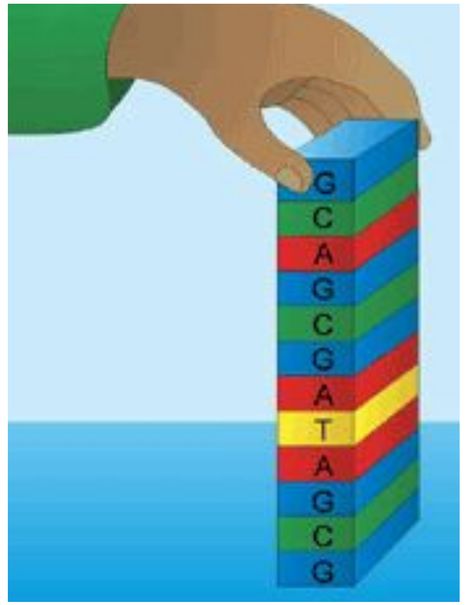
Paramecium
(草履蟲)

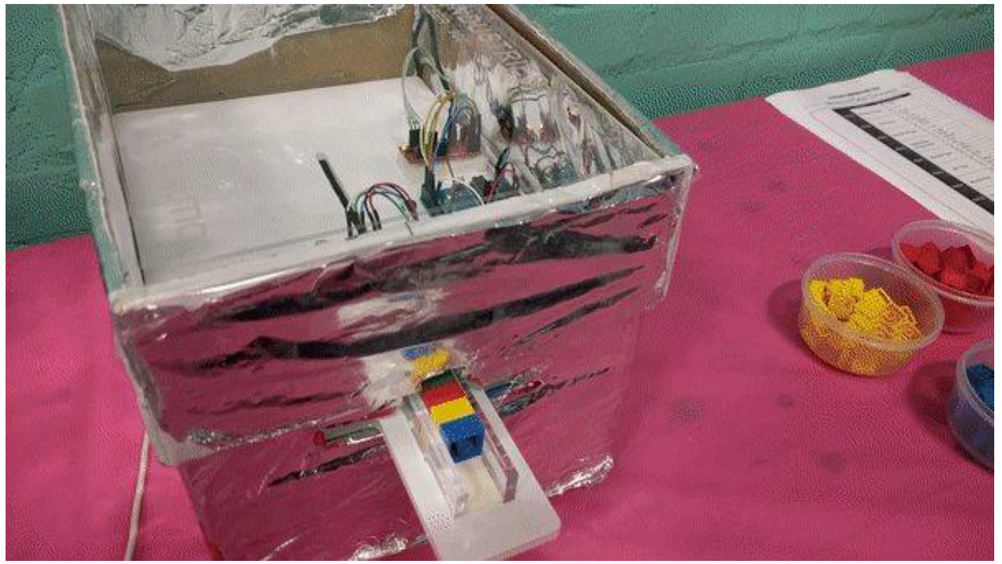


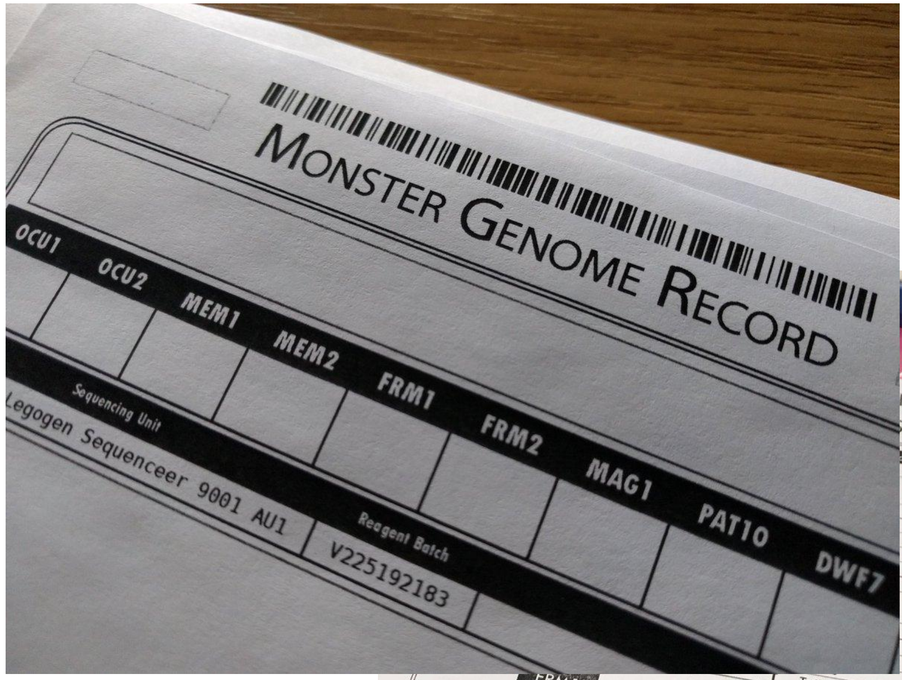
Amoeba (變形蟲)

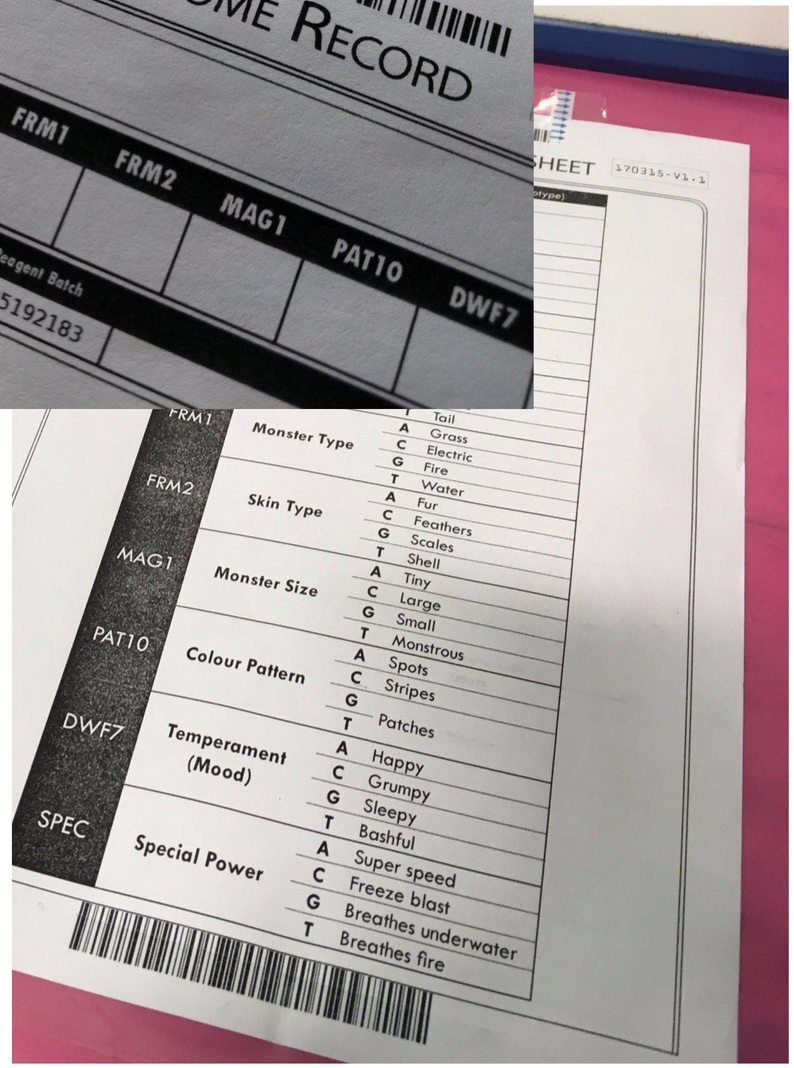
Lego DNA Sequencer

Using Lego as analogy of 4 DNA nucleotides
- ATCG

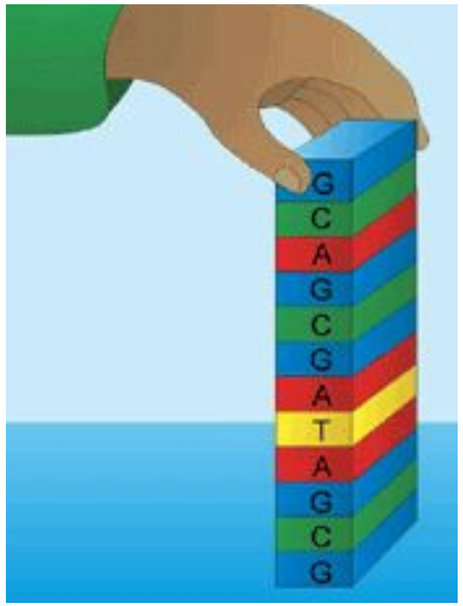




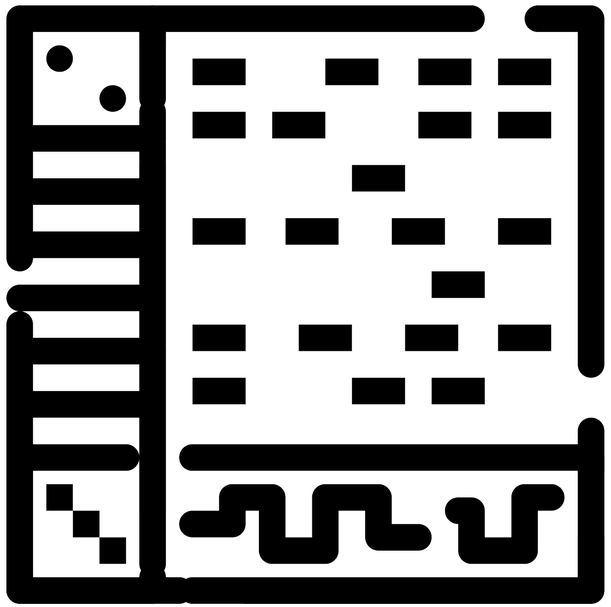




Lego DNA Sequencer



DNA Stack

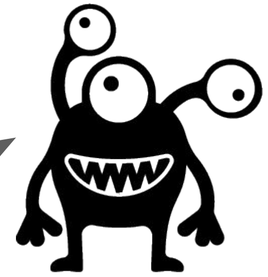


Sequencer

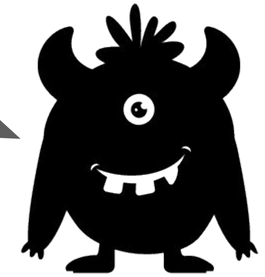
No. of eyes?

Mood?

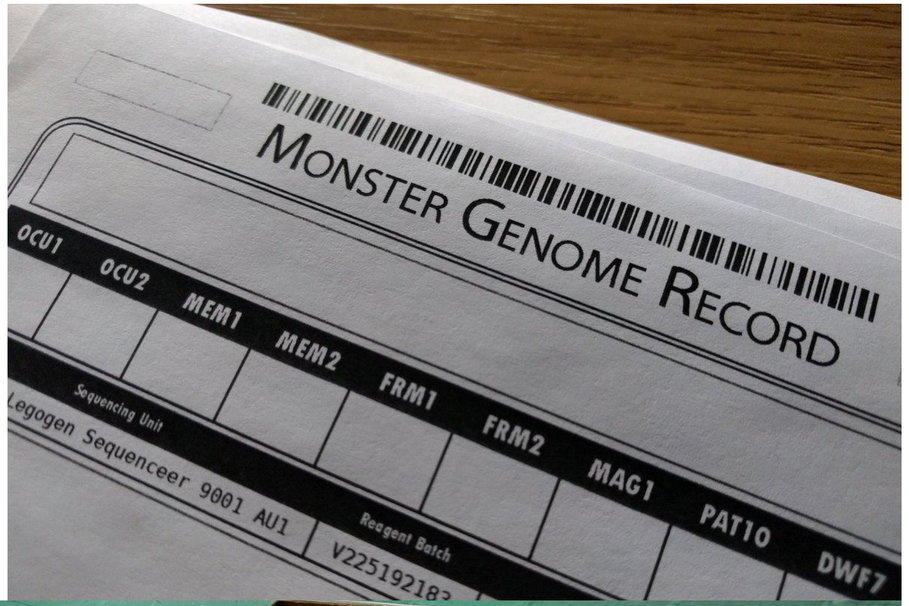
Special Power

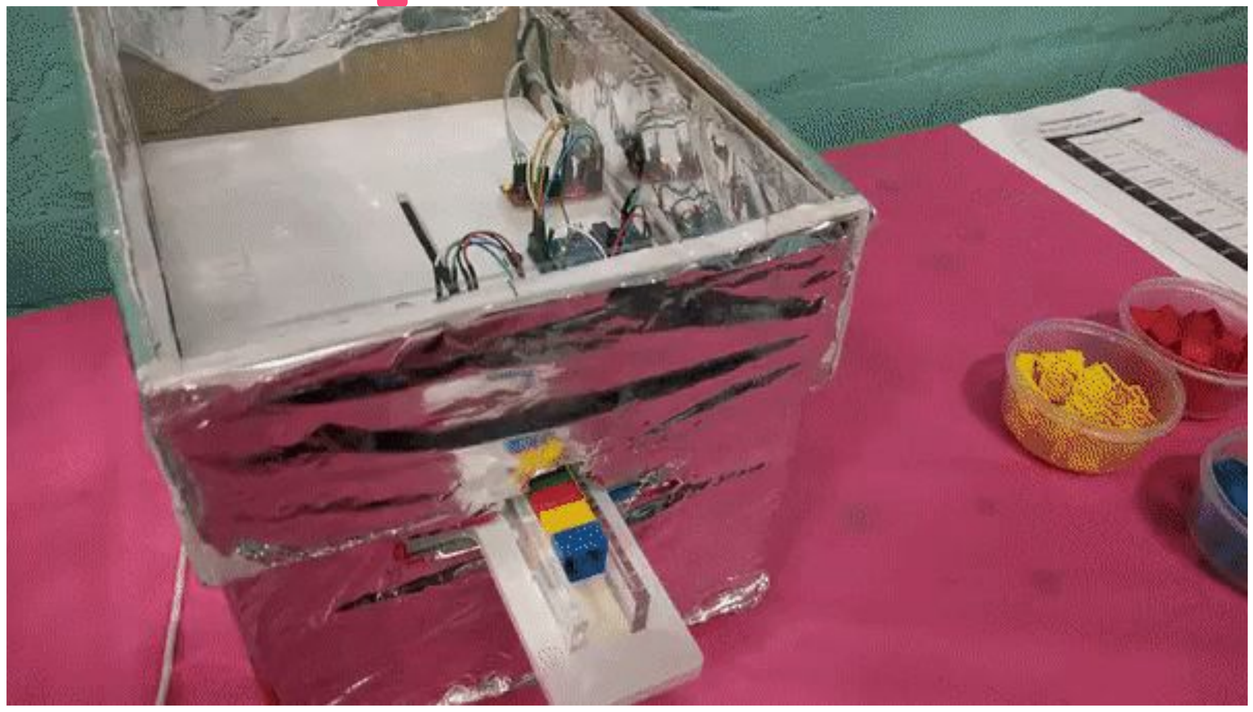






Lego DNA Sequencer







UV-010-01

Monster Gene Datasheet

170315-V1.1

	Description	Base	Result (Phenotype)
OCU1	**Number of Eyes**	**A**	1 Eye
		C / **G**	2 Eyes
		T	8 Eyes
OCU2	**Eye Colour**	**A**	Blue
		C	Green
		G	Brown
		T	Random
MEM1	**Number of Legs**	**A**	2 Legs
		C / **G**	4 Legs
		T	8 Legs

MEM2	**Appendages or Extremities**	**A**	Wings
		C	Antennae
		G	Horns
		T	Tail
FRM1	**Monster Type**	**A**	Grass
		C	Electric
		G	Fire
		T	Water
FRM2	**Skin Type**	**A**	Fur
		C	Feathers
		G	Scales
		T	Shell
MAG1	**Monster Size**	**A**	Tiny
		C	Large
		G	Small
		T	Monstrous

PAT10	**Colour Pattern**	**A**	Spots
		C	Stripes
		G	Patches
		T	
DWF7	**Temperament (Mood)**	**A**	Happy
		C	Grumpy
		G	Sleepy
		T	Bashful
SPEC	**Special Power**	**A**	Super speed
		C	Freeze blast
		G	Breathes underwater
		T	Breathes fire

30+ Experiment

- 6 - 18 y.o.
- Apparatus + Consumables
- Life Sci. & Biotech (1st phase)
- Engineering & Art (2nd)





Astrobiology

Marine Science

Micro fluidics

Water Recycle

DNA Extraction & Sequence

Renewable Energy from Plants

Kitchen Science

Sensation of Living Things

Microworld

Protection of Organs

Bionic Robot

Hygiene

Waste Water Treatment

Ecology

Vital Signal

Cooking by Sun

Organ on Chip

Metabolism



more experiments...

From Washing Hand to DNA Editing Experiment



Bioart

1. **Isolate** soil microbes
2. **Identify** microbes
3. Make **agar art** (paintings using microbes)



4. Create hypothesis **based on art**
5. **Experiment**
6. Present findings

DIY Plant Based Meat



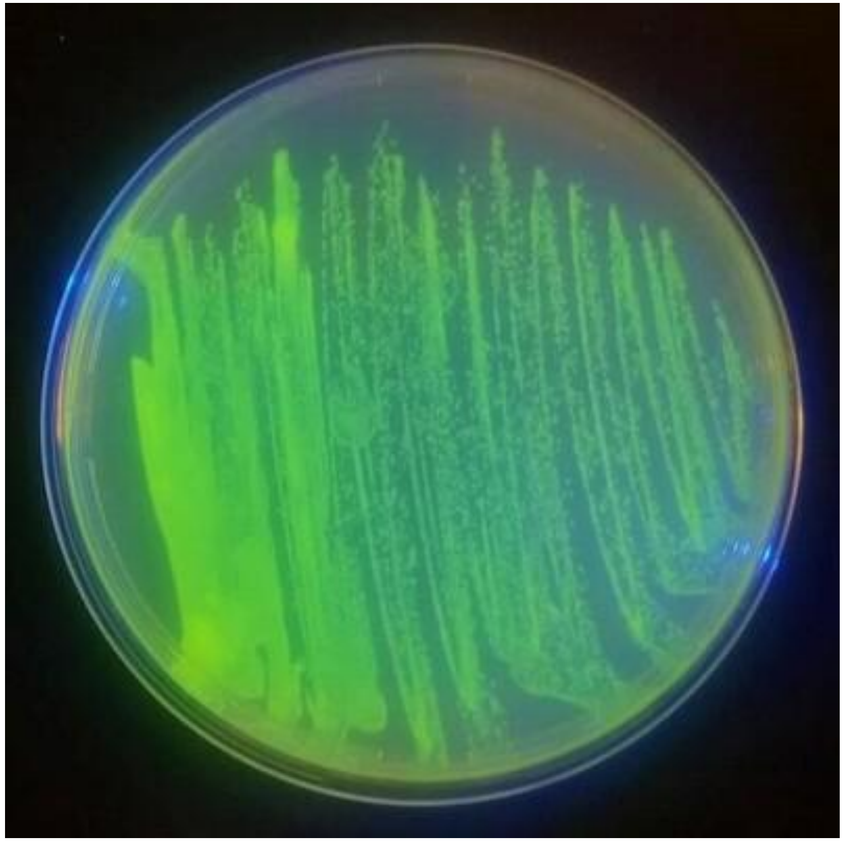
Figure 1: Raw vegan burger mixture with **Metolose® MCE-100TS**



Figure 2 and 3: Fried vegan burger patty with **Metolose® MCE-100TS**

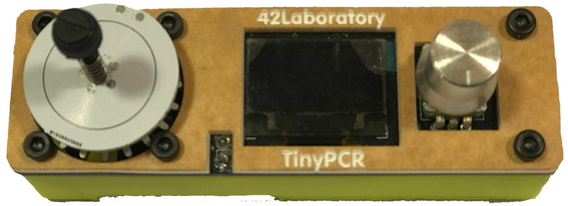


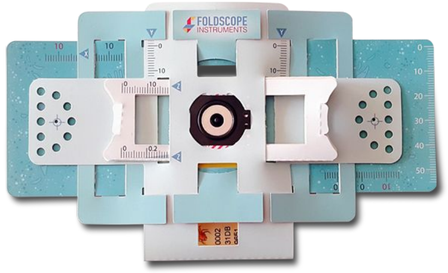
Green Fluorescent Protein(GFP) -jelly fish

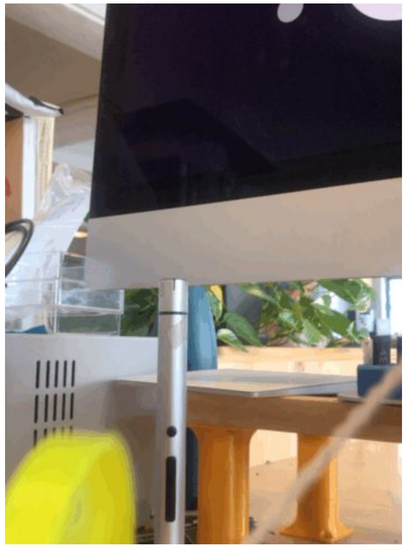


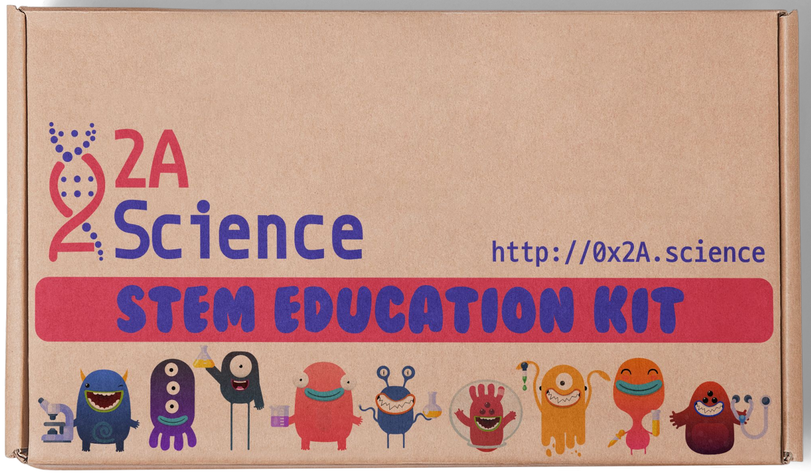
2A Science

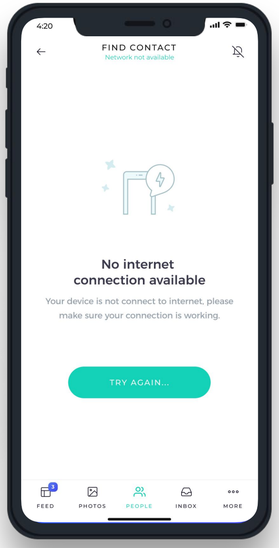
http://0x2A.science

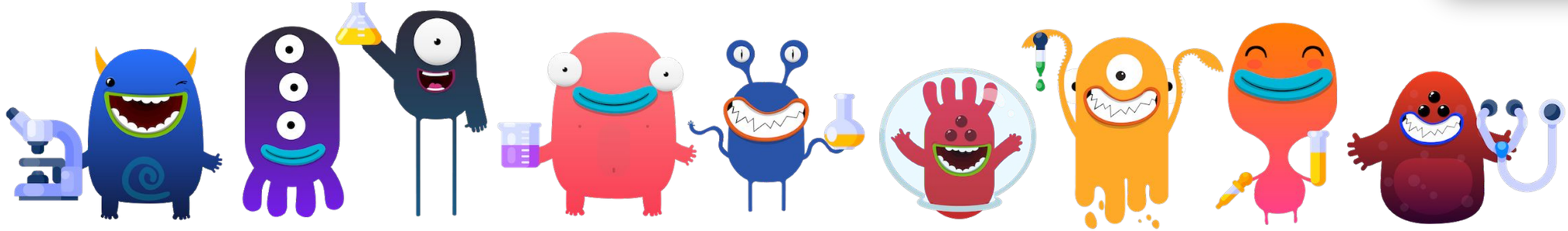










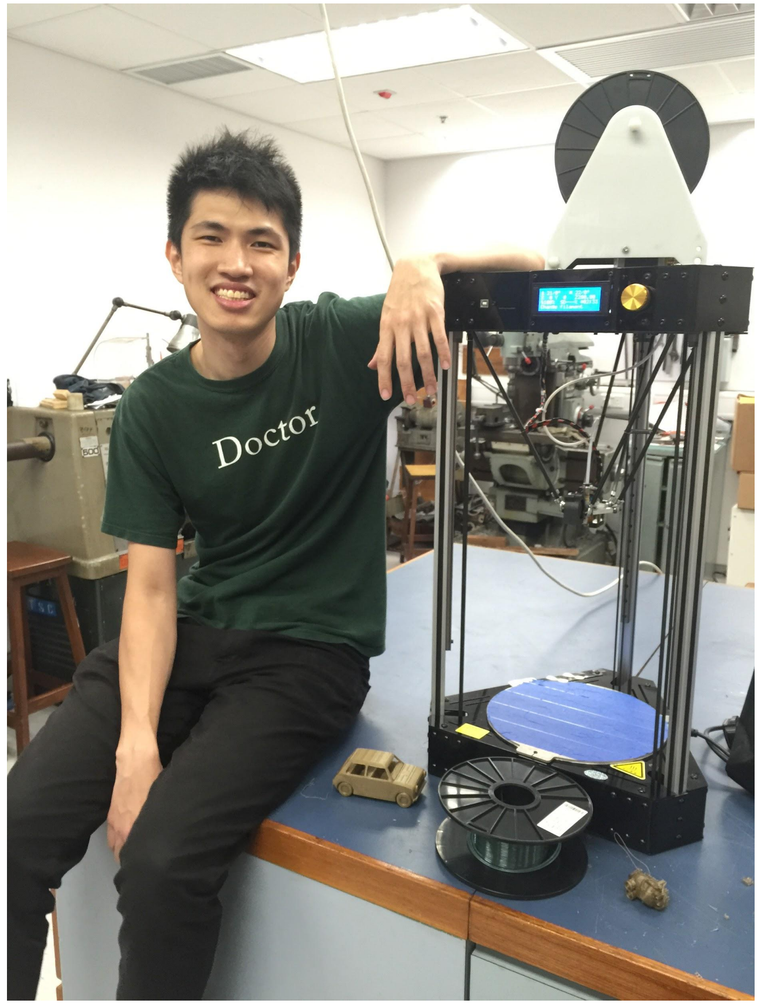


Engineering

Maker Academy

DO SOMETHING THAT MATTER

- BEng in Electronics Dropout
- Experimental Physicist specializing in biophysics, Positron Physics and Semi-conductor Physics

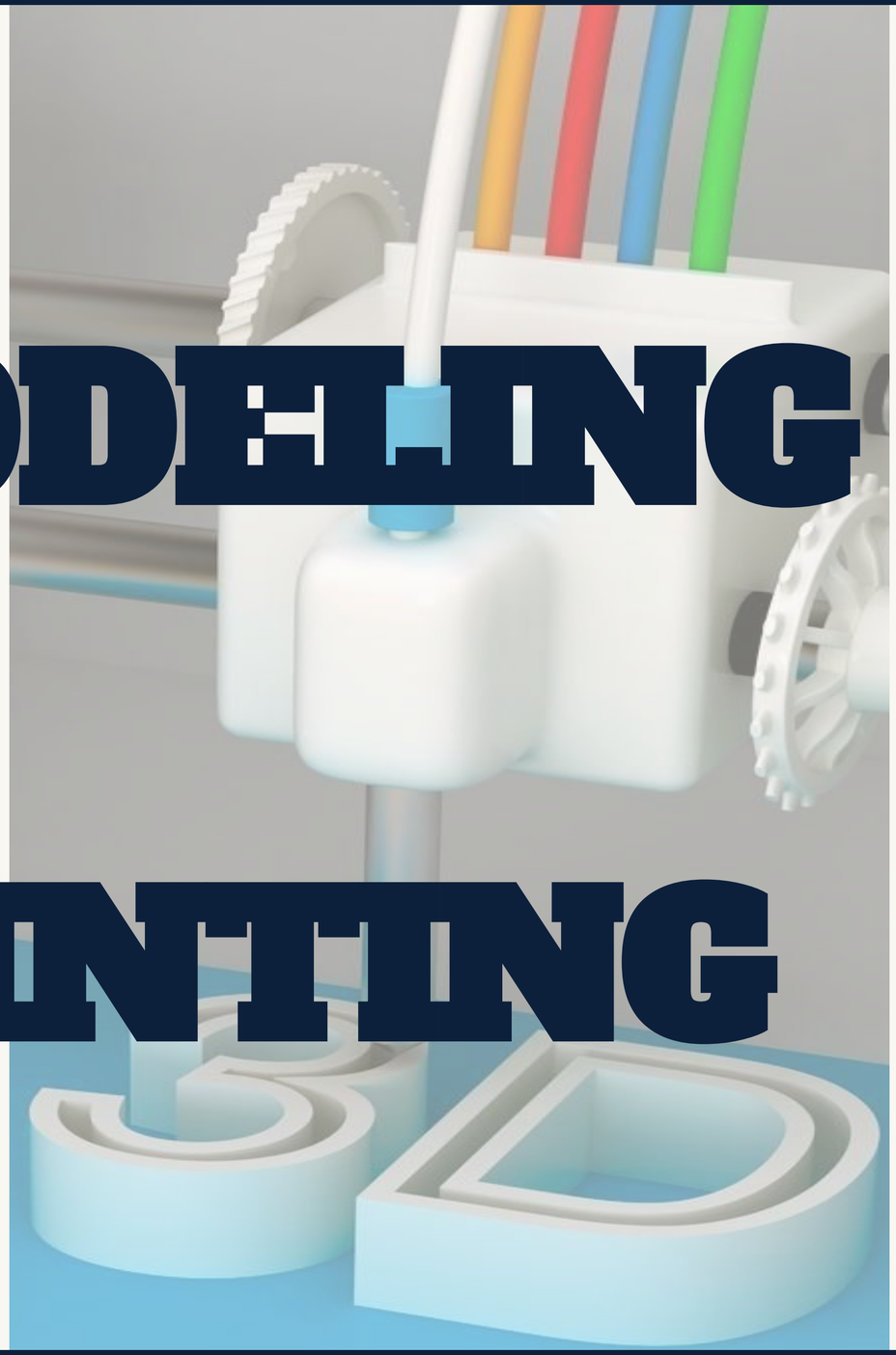


INTRO. TO

3D MODELING

&

3D PRINTING



CONTENT

- Principles of 3D Printing
- Introduction to Computer Aided Design(CAD) tools for 3D modelling
- 3D Printing Tools

REQUIREMENT

- Internet connected laptop with mouse and latest Chrome Browser
- Validated email for Autodesk services

DURATION

- 1-hr talk + 1-hr workshop

Maker Academy

DO SOMETHING THAT MATTER

INTRO. TO CAD DESIGN & LASER CUTTING



CONTENT

- Principles of Laser Cutting
- Introduction to Computer Aided Design(CAD) tools for laser cutting

REQUIREMENT

- Internet connected laptop with mouse and latest Chrome Browser
- Validated email for autodesk services

DURATION

- 30-min lecture + 1-hr workshop

Maker Academy
DO SOMETHING THAT MATTER

INTRO. TO CAD DESIGN & CNC CUTTING



CONTENT

- Principles of Computer Numerical Control (CNC) Cutting
- Introduction to Computer Aided Design(CAD) tools for 2.5D modelling

REQUIREMENT

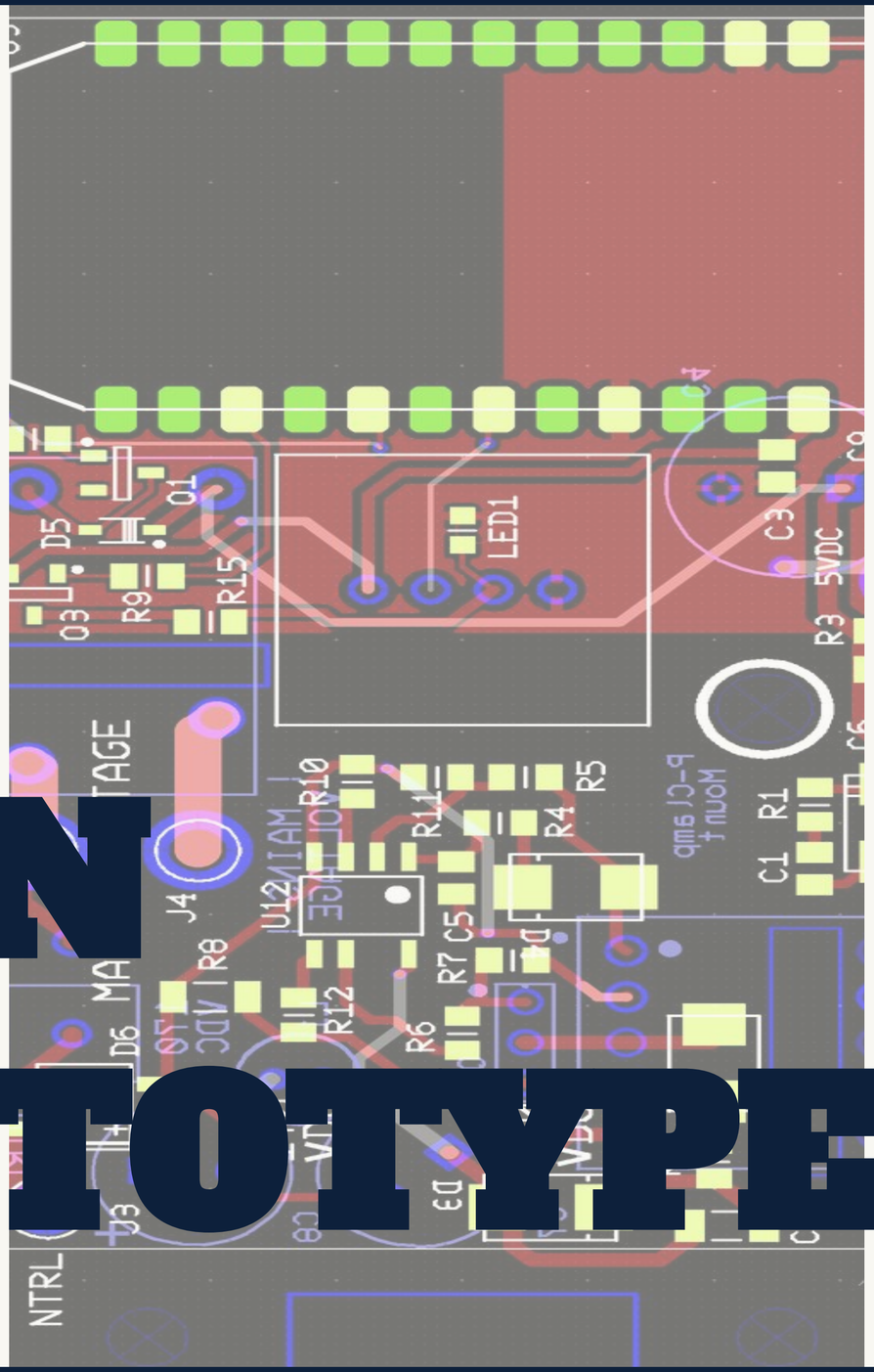
- Internet connected laptop with mouse and latest Chrome Browser
- Validated email for autodesk services

DURATION

- 30-min lecture + 45-hr workshop

Maker Academy
DO SOMETHING THAT MATTER

INTRO. TO BASIC PCB DESIGN & PROTOTYPE



CONTENT

- Basic Electronic Principle
- Basic tools for electronic simulation and prototyping

REQUIREMENT

- Internet connected laptop with mouse and latest Chrome Browser
- Validated email for autodesk services

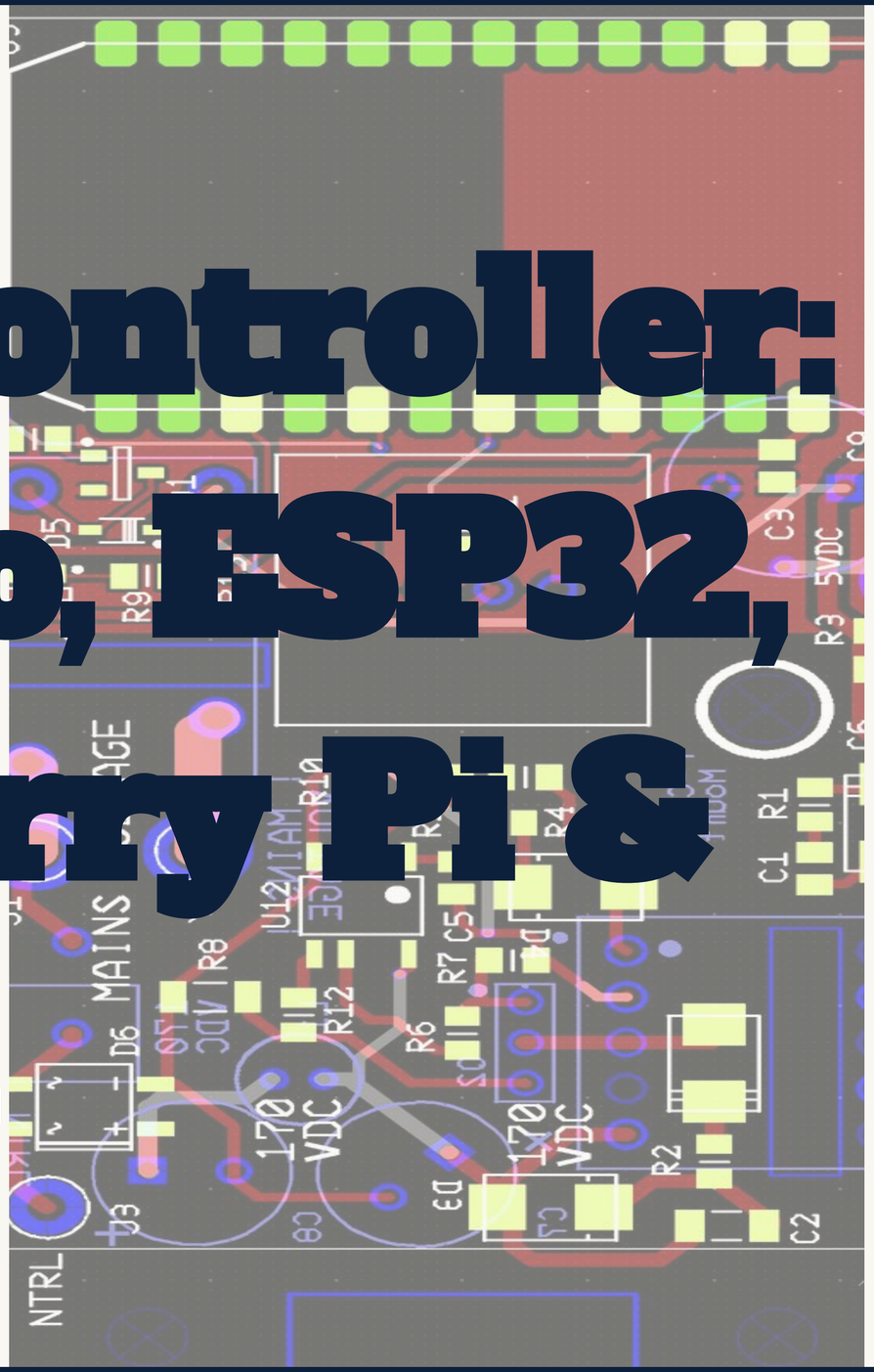
DURATION

- 45-min lecture + 45-min workshop

Maker Academy
DO SOMETHING THAT MATTER

INTRO. TO

Microcontroller: Arduino, ESP32, Raspberry Pi & Beyond



CONTENT

- Basic Electronic Principle
- Basic working principle of microcontrollers
-

REQUIREMENT

- Internet connected laptop with mouse and latest Chrome Browser
- Validated email for autodesk services

DURATION

- 45-min lecture + 45-min workshop

Maker Academy
DO SOMETHING THAT MATTER

INTRO. TO GRAPHIC DESIGN WITH CANVA



CONTENT

- Elements and effective communication for graphic design
- Using Canvas for marketing tools

REQUIREMENT

- Internet connected laptop with mouse and latest Chrome Browser
- Validated email for Canva services

DURATION

- 30-min lecture + 1-hr workshop

Maker Academy
DO SOMETHING THAT MATTER

INTRO. TO UI DESIGN & MOCKUP FOR MOBILE APP



CONTENT

- Basic of UI Design
- Tools for wire framing and prototyping for mobile application

REQUIREMENT

- Internet connected laptop with mouse and latest Chrome Browser
- Validated email for wire-framing and prototyping tools & services

DURATION

- 45-min lecture + 1-hr workshop

Maker Academy
DO SOMETHING THAT MATTER

Computation

|0>+|1>
School

- Technology Coach
- 1st Qiskit / Quantum Computing Advocate of IBM in HK
- DeFi (Decentralised Finance) HK Organizer
- Host of Blockchain Radio Programme



7th Aug 2020

DeFi Meet-up (Zoom) at MIT Innovation node



Aug - Oct 2019

區塊鏈與你電台節目

INTRO. TO

CODING WITH BLOCKY





CONTENT

- Basic logic of programming
- Making a mini project

REQUIREMENT

- Internet connected laptop with mouse and latest Chrome Browser

DURATION

- 45-min lecture + 1-hr workshop

Deeptech School

INTRO. TO

Low/ No Code App or AI Development



CONTENT

- Basic logic of programming
- Making a mini project

REQUIREMENT

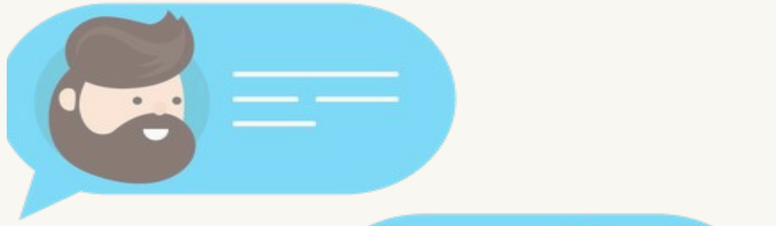
- Internet connected laptop with mouse and latest Chrome Browser

DURATION

- 45-min lecture + 1-hr workshop

Deeptech .School

INTRO. TO DATA SCIENCE & MACHINE LEARNING









CONTENT

- Basic knowledge of Data Science
- Basic knowledge of Machine Learning
- Types of Machine Learning Algorithms

REQUIREMENT

- Internet connected laptop with mouse and latest Chrome Browser
- Validated email for Microsoft and Google services

DURATION

- 45-min lecture + 1-hr workshop

Deeptech School

INTRO. TO DEEP LEARNING & AWS DEEP RACER



CONTENT

- Basic knowledge of reinforced learning
- Introduction of Deep Racer of AWS

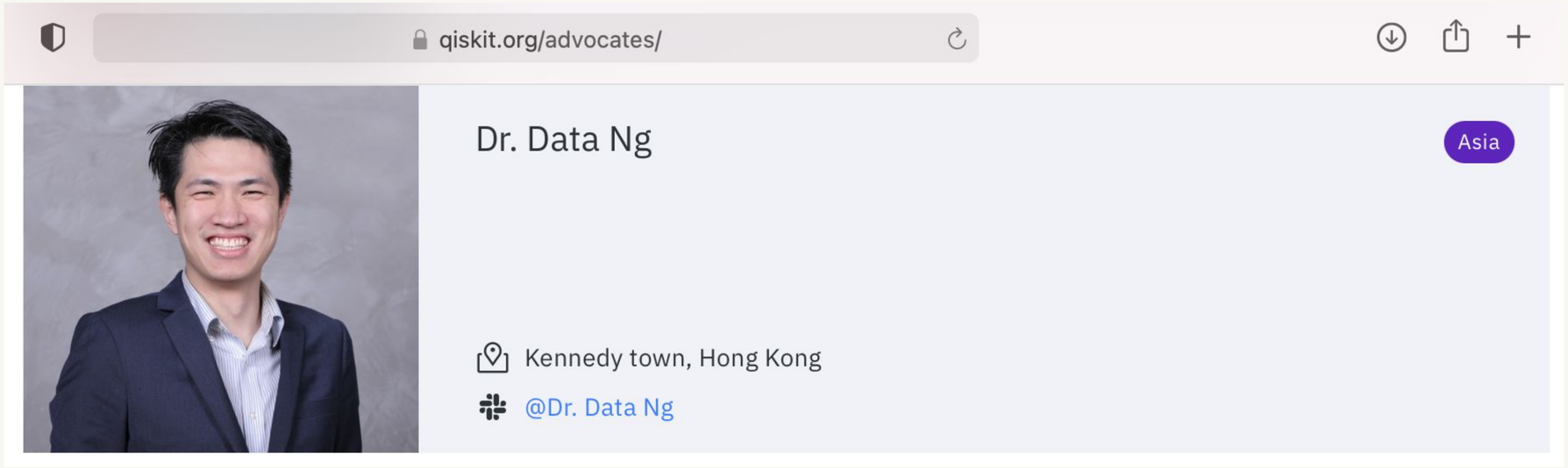
REQUIREMENT

- Internet connected laptop with mouse and latest Chrome Browser
- Validated email for AWS services

DURATION

- 45-min lecture + 1-hr workshop

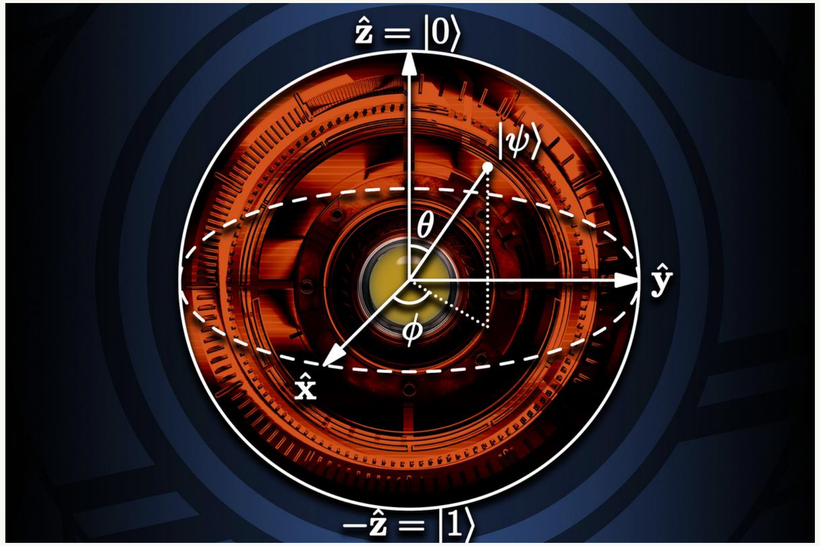
Deeptech .School

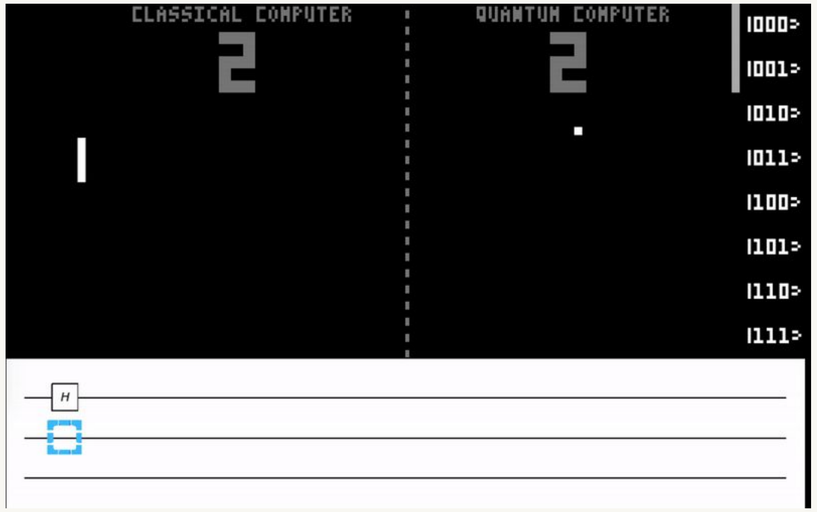


- First Qiskit Advocate in HK / GBA appointed by IBM

INTRO. TO

Quantum Computing





CONTENT

- Basic knowledge of Quantum Computing
- Introduction of Qiskit
- Quantum Simulation

REQUIREMENT

- Internet connected laptop with mouse and latest Chrome Browser

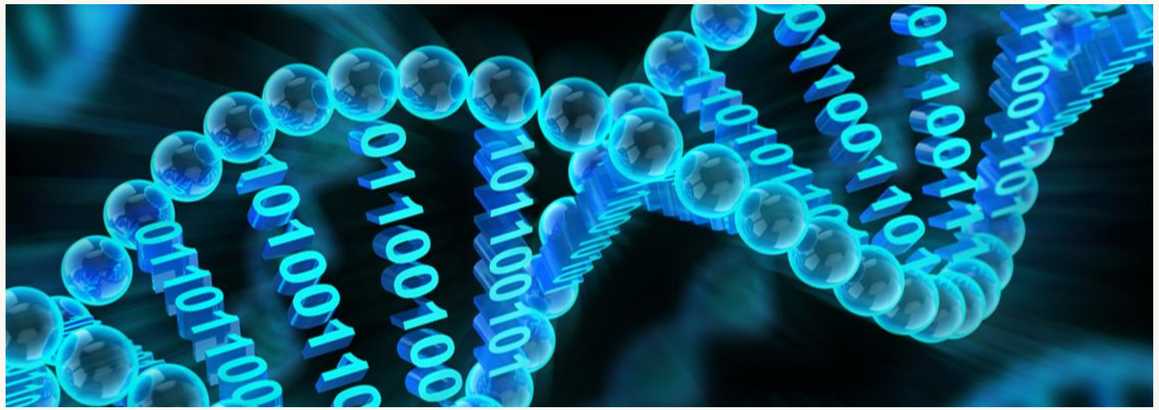
DURATION

- 45-min lecture + 1-hr workshop

Deeptech .School

INTRO. TO

Bioinformatics



CONTENT

- Basic knowledge of bioinformatics
- Application of bioinformatics

REQUIREMENT

- Internet connected laptop with mouse and latest Chrome Browser

DURATION

- 45-min lecture + 1-hr workshop

Deeptech .School

INTRO. TO

DeFi

& Blockchain

CONTENT

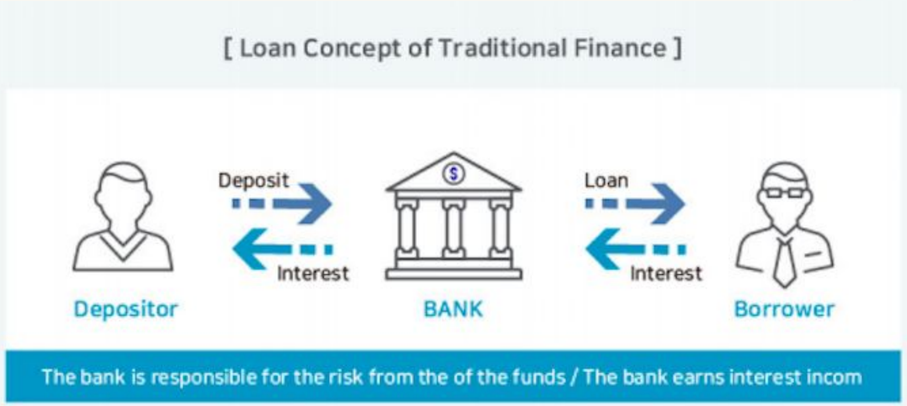
- Basic knowledge of DeFi & Blockchain
- Application of Blockchain
- Importance of DeFi

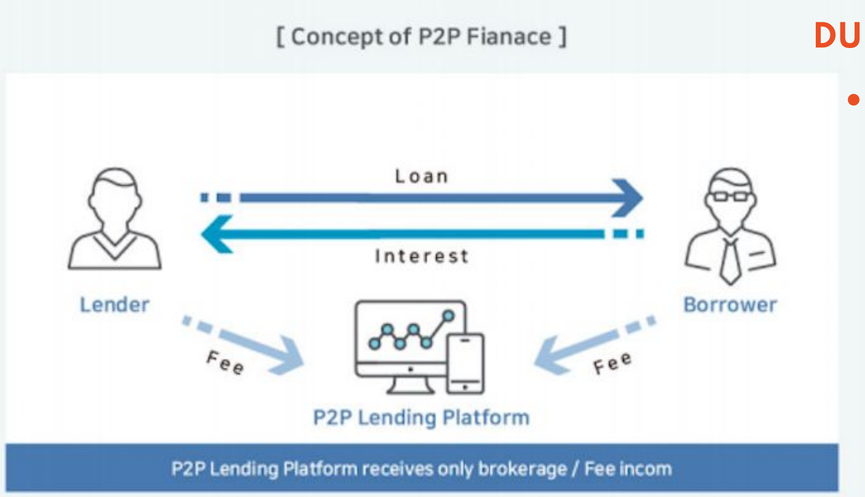
REQUIREMENT

- Internet connected laptop with mouse and latest Chrome Browser

DURATION

- 45-min lecture + 1-hr workshop





Deeptech School

INTRO. TO

Tokenization & Token Economy



CONTENT

- Basic knowledge of Blockchain, Mining, PoW & PoS, Tokenization, Fungible and non-fungible tokens, and Token Economy
- Application of Tokenization
- Play-to-earn
- Importance of Token Economy

REQUIREMENT

- Internet connected laptop with mouse and latest Chrome Browser

DURATION

- 45-min lecture + 1-hr workshop

Deeptech School

http://0x2A.science Data@0x2A.science









